

Usman Ahad

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EDUCATION

Lahore University of Management Sciences

Bachelor of Science in Computer Science, CGPA: 3.69 / 4.00

Lahore, Punjab, Pakistan

Aug. 2023 – Present

- Dean's Honour List for 3 consecutive academic years; graduate coursework in Advanced Topics in Machine Learning, Deep Vision-Language Models, AI on Edge Devices, Deep Learning, and Internet of Things

RESEARCH EXPERIENCE

VisConf: Quality-Aware Consensus for Best-of- N VLM Reasoning

PyTorch, Transformers, Qwen2.5-VL, InternVL3, Gemma 4

Jan. 2026 – Sep. 2026

- Developed a training-free VLM selection framework combining candidate-calibrated Self-Certainty, visual engagement, and Rank-Weighted Consensus without external verifier or reward models
- Achieved **56.36%** mean accuracy across 3 VLMs, 3 benchmarks, and $N \in \{8, 16, 32\}$; outperformed all baselines in **24 of 27** settings and Self-Consistency by 0.98 points

Mind Your Own Business: Consent-Based Audio Privacy for Smart Glasses

PyTorch, Qwen3.5-2B, LoRA, DPO, RAG, LS-EEND

Jan. 2026 – Sep. 2026

- Designed a semi-real-time Raspberry Pi 5 pipeline for diarization, streaming transcription, cross-window disclosure tracking, speech anonymization, and consent-gated restoration of original audio
- Distilled GPT-5.6 Luna into Qwen3.5-2B using 30,960 SFT examples and 6,241 DPO pairs; reduced leakage by **24.7%** while raising utility by 2.5 points on 143 held-out transcripts

FedCPR: Federated Open-Set Recognition with Latent Prototype Rejection

PyTorch, Federated Learning, Open-Set Recognition

Nov. 2025 – Feb. 2026

- Developed latent-space outlier synthesis with curriculum learning under FedAvg, achieving **85.42% AUROC** and 90.96% closed-set accuracy on CIFAR-10 across 5 non-IID clients

WatchTower: Transformer-Based Network Anomaly Detection for Edge Devices

PyTorch, TFLite, Apache TVM, FedProx, PROSER, Raspberry Pi 5

Aug. 2025 – Dec. 2025

- Built a memory-token Transformer with PROSER latent outliers and FedProx, achieving **93.66% accuracy** and 92.83% F1 on CIC-IDS2017 with live DoS validation
- Compressed the model by **67.5%** and used 1,000-trial TVM auto-tuning to reach 3.82 ms single-core latency and 260+ packets/sec, a $2.6\times$ speedup

TEACHING EXPERIENCE

Teaching Assistant, AI on Edge Devices

Lahore University of Management Sciences

Aug. 2026 – Present

Lahore, Punjab, Pakistan

- Support graduate instruction in measurement, quantization, hardware-aware compilation, TinyML, edge LLMs, and federated privacy across microcontroller and edge-accelerator targets

Teaching Assistant, Deep Learning

Lahore University of Management Sciences

Jan. 2026 – May 2026

Lahore, Punjab, Pakistan

- Supported 100+ MS AI students through assignments, office hours, and evaluation across transformers, generative models, and diffusion models

TECHNICAL SKILLS

ML Frameworks & Systems: PyTorch, Transformers, Hugging Face, ONNX Runtime, TFLite, Apache TVM, Scikit-learn, Docker, Linux

Research Areas: Vision-Language Models, Test-Time Scaling, Audio Privacy, Preference Optimization, Federated Learning, Open-Set Recognition, Quantization, Hardware-Aware Compilation

Languages & Platforms: C/C++, Java, SQL, JavaScript, MATLAB, Raspberry Pi 5, Edge Hardware